



Sofie—the surgical robot

the world to control it) but also haptic feedback. Haptic feedback enables the surgeon controlling Sofie to 'feel' the resistance from the patient's tissues in response to his actions. The sense of touch is emulated through joysticks, which become harder to move when there is more pressure from the tissues. So the surgeon feels the impact as he remotely performs the surgery using joysticks.

The olfactory system, or the sense of smell, has also been very difficult to replicate in the past. In recent years, gas sensors and other sensors that detect the constituents of the surrounding air hand-in-hand with software that interpret these into their characteristic 'smells' have resulted in the development of some basic models of electronic noses.

Robots can also talk, although the ability of robots to converse has been used mainly for entertainment purposes. Talking has been helpful in certain healthcare applications where robots respond verbally to patients who cannot see visual responses or offer soothing words to ailing people. Robots have, however, long been intelligent enough to engage in long conversations, whether verbally or through electronic text. Remember KISMET—the Massachusetts Institute

of Technology (MIT) heart-throb that could chat online?

There have been some attempts at bringing the sixth sense too into a robot. "Feeling has been simulated in humanoid robots like the Geminoid, by receiving electrical signals from the person it is being modelled upon. Other emotions and human feelings are implemented mainly by the use of artificial neural networks. The concept of neural networks has existed for long, and it is now being used in the software for robots," says Pitchai.

Dr G Venkatesh, who heads Bangalore Robotics (a non-profit hobby and research centre based in Bengaluru), says that the sensing abilities of today's robots are, in some respects, even better than humans'.

"There are all kinds of sensors available now, thanks to micro-electromechanical systems (MEMS) and nanotechnology. A robot can be equipped with a whole range of integrated sensors—something that is non-existent in natural life forms. This includes sensor aggregation, where a multitude of sense organs can be fabricated as a single, independent, functional unit. In some ways, humans and other life forms have a limited sense-perception mechanism. For example, we can sense distance but can rarely be accurate in

its measurement. Artificial sensors can be quite accurate in this regard. This is possible because every sensor can be built with its own 'brain.' The result, if integrated into a robot, is a vastly superior sensory system," says Dr Venkatesh.

Tech boosting robots' senses

The advanced sensing ability of today's robots is both due to new technologies as well as innovative applications of existing technologies. Pitchai cites an example: "The wine-tasting robot developed by NEC Technologies is one that uses infrared spectroscopic technology, which has existed for a very long time. Of course, the sensors were made to be more accurate and sense a wider spectrum that ranges from near-infrared rays to the mid-infrared spectrum."

Here is a quick review of some of the technologies that have contributed to the current state of robotic perception.

Micro-electromechanical systems. MEMS technology has allowed integration of several sensor functions into a compact package. Using individual sensors for each requirement increases the cost, footprint and number of wires to be managed. Putting it all into a single package using MEMS technology makes the job much simpler for robot makers.

Another significant contribution of MEMS technology is inexpensive vibrating-structure gyroscopes, which provide a feedback-sensing mechanism that can be very useful in improving the orientation and autonomous navigation capabilities of robots. MEMS gyroscopes are packaged like integrated circuits, and often a single unit includes gyroscope sensors for multiple axes. If the gyroscope is combined with an accelerometer as well, the output has six full degrees of freedom.

Nanotechnology. The relationship of nanotechnology with robotics can be seen at two scales—present and future. In the future, nanotechnology will re-